



SAFETY



PRODUCTION



COST



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PIT DEWATERING CONTROL SYSTEM

27/01/2026

exxaro

GROOTEDELUK | COAL



- ➔ GGC Onboarding and Exposure
- ➔ Project
- ➔ Learning Outcomes

Week 1 & 2: Onboarding

- Medicals
- Inductions
- Access Card
- Engineering Development

Week 3:

Given Project & stationed at GG78
Technical Services (TS) Offices

Week 4:

Project & TS exposure

Week 5 & 6:

Refining project

Problem Description

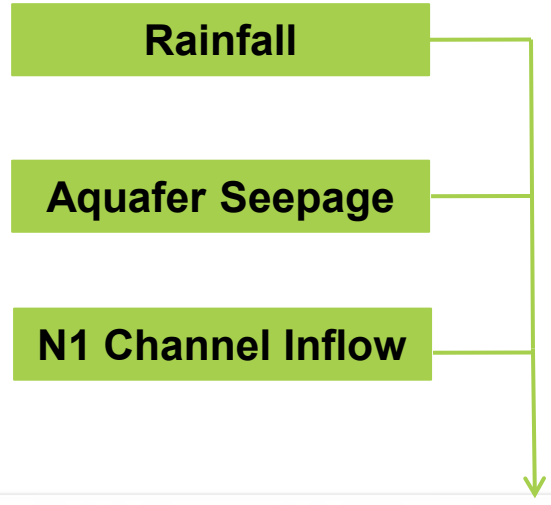


Climate Change

[ˈklī-mæt ˈchānj]

The long-term shift in global and local weather patterns primarily caused by carbon emissions and the greenhouse effect.

Image From: [ecolife.com](https://www.ecolife.com)



Observation	• The mining sector is impacted by climate change • GGC's pit has accumulated an immense amount of water
Problems	• Affects Mining Operations • Affects Production • Risk of backfill material displacing the water further north



 Pit Water Management Strategy

GGC Pit (20/01/2026)

Problem Statement

Investigate a Smart Drive System (SDS) with Advanced Process Control (APC) to automate pit dewatering, to reduce Scope 2 emissions, and protect mining assets.



Scope 2 emissions: The emissions released into the atmosphere from the use of purchased energy that is generated at another facility such as a power station. (McClymont, 2021)



Smart Drive System: An integrated motor control solution that leverages built-in intelligence and connectivity to transition from reactive to predictive operations (Schneider Electric, 2023; Siemens Global, 2024).

Current Control Philosophy

Currently the Yellow Pipeline is designed to operate on a Regulated Control philosophy using Direct-on-Line (DOL) starters.

Problem

- Exposes pipeline to hydraulic surges that lead to mechanical wear
- Inefficient power consumption
- High Total Cost of Ownership



Blue Pipeline B6 Pump and Motor

Direct Coupling

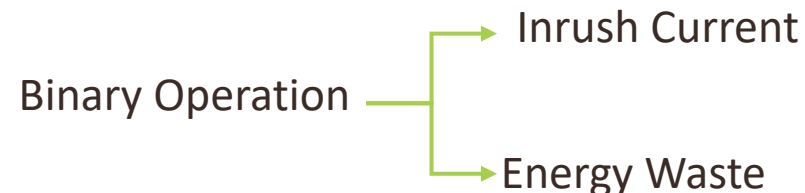
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Proposal:

1. Incorporate Variable Speed Drives (VSD)
2. Incorporate Pressure Transmitters along the pipeline
3. Incorporate Advanced Process Control (APC)



VSD Proposal



Image From: weg.net

Feature	Description
Model	Weg CFW11
Voltage [V]	600
Current [A]	260
Power [kW]	280
Efficiency	≥ 97%
Price per unit	R950 000

Motor Proposal

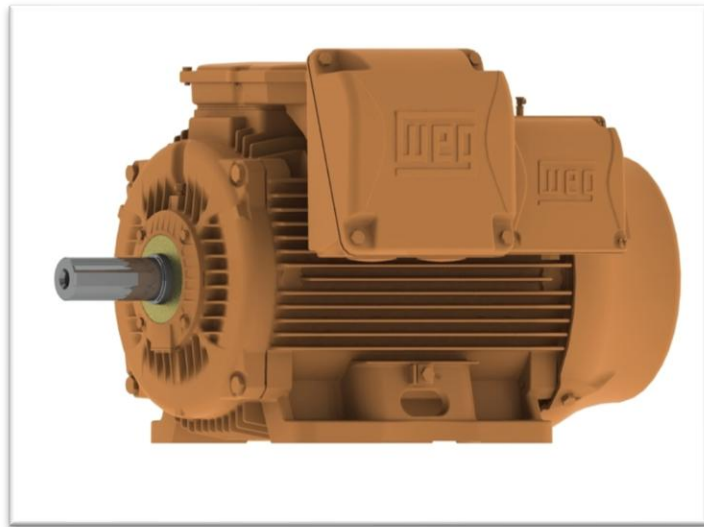


Image From: weg.net

Feature	Description
Model	Weg W22 IE2
Voltage [V]	600
Current [A]	248
Power [kW]	250
Speed [rpm]	1490
Efficiency	95.7%

Case 1

Costs of running the pipeline at 100% speed using direct coupling for 24 hours a day over one year.

Parameter	Formula	Value
Electric Power (kW)	$\frac{\text{Mechanical Power}}{\eta_{motor}} \times 4$	1044.93 kW
Energy (kWh)	$\text{Electric Power} \times \text{time}$	25 078.37 kWh
Daily cost of electricity	$\text{Energy} \times \text{Electricity Rate } (\frac{R}{kWh})$	R27 586,21
Yearly cost of electricity		R 10 068 965.52

Electricity Rate for Industrial (MegaFlex): R0.50 – R1.10

Case 2

Costs of running the pipeline at 80% speed using VSDs for 24 hours a day over one year.

Parameter	Formula	Value
Speed at 80%	$RPM \times 80\%$	1192 rpm
Mechanical Power (kW)	$\left(\frac{Speed\ at\ 80\%}{Speed\ at\ 100\%} \right)^3 \times Rated\ Mech\ Power$	128 kW
Electric Power (kW)	$\frac{Mechanical\ Power}{\eta_{motor} \times \eta_{VSD}} \times 4$	551.55 kW
Energy (kWh)	$Electric\ Power \times time$	13 237.24 kWh
Daily cost of electricity	$Energy \times Electricity\ Rate\ (\frac{R}{kWh})$	R14 560.97
Yearly cost of electricity		R 5 314 745.05

Electricity Rate for Industrial (MegaFlex): R0.50 – R1.10

Consolidation

Total Energy saved in one year: 4 322 012.45 kWh

Financial Component	Value	Contribution to Payback
Gross Investment	(3 800 000)	Total CAPEX for four VSDs
Project Overhead	(1 570 000)	Estimated Overheads for the year
Estimated Project Cost	(5 370 000)	Costs for the project in the first year
Year 1 Energy Saving	+4 754 220	Recurring OPEX reduction
Year 1 Tax Rebate	+1 108 596	One-time Section 12L tax rebate
Net Year 1 Position	+ 492 816	Recovers 100% of capital for the VSDs 12 months

Explanation

Due to Affinity Law 3, we will achieve an ROI within the first year of commissioning the project.

$$P_{80\%} = \left(\frac{N_{80\%}}{N_{100\%}} \right)^3 \times P_{100\%}$$

The APC

- An intelligent control layer which manages coordinated systems.
- It aids in calculating the pipeline's theoretical hydraulic limits then conducts a series of calculations to set the individual speed setpoints of each pump such that the entire system reaches a desired flow rate.

APC Benefits:

- Prevents Human Error
- Increases safety
- Increases machine lifespan
- Lower operating costs

APC key techniques:

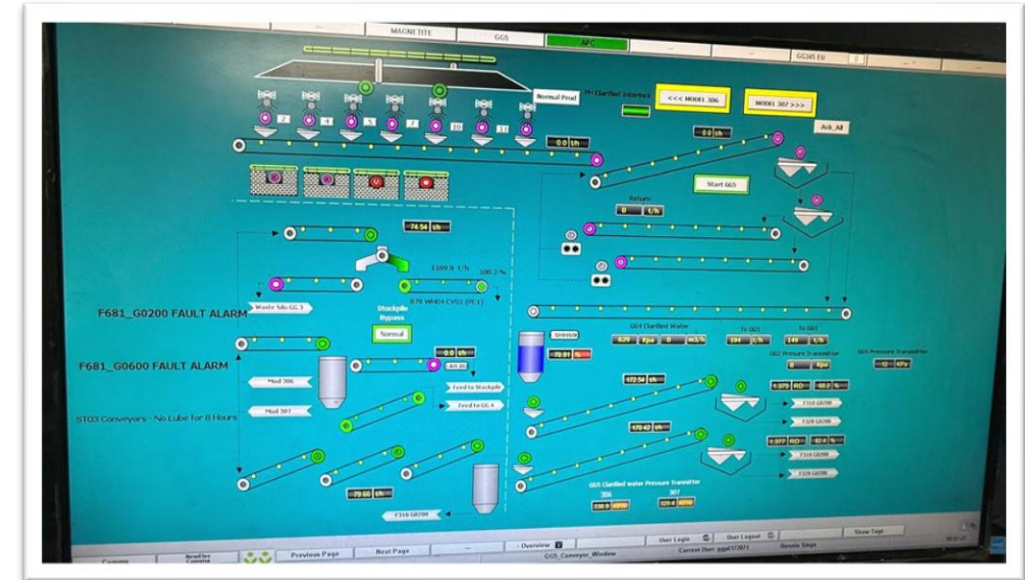
- Model Predictive Control
- Multivariable Control
- Digital Twin
- Optimization algorithms

APC



GG5_APC_Window

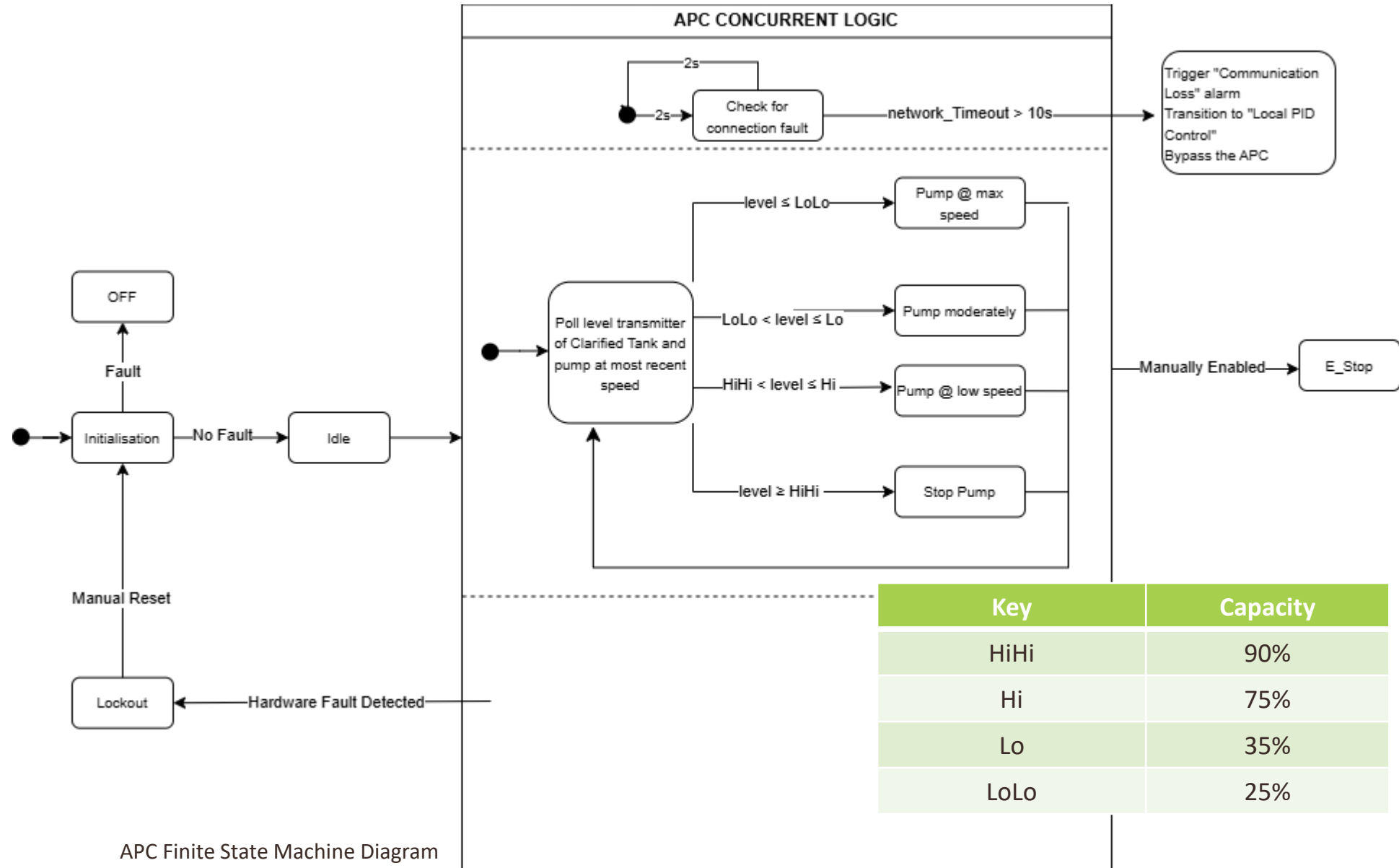
SCADA

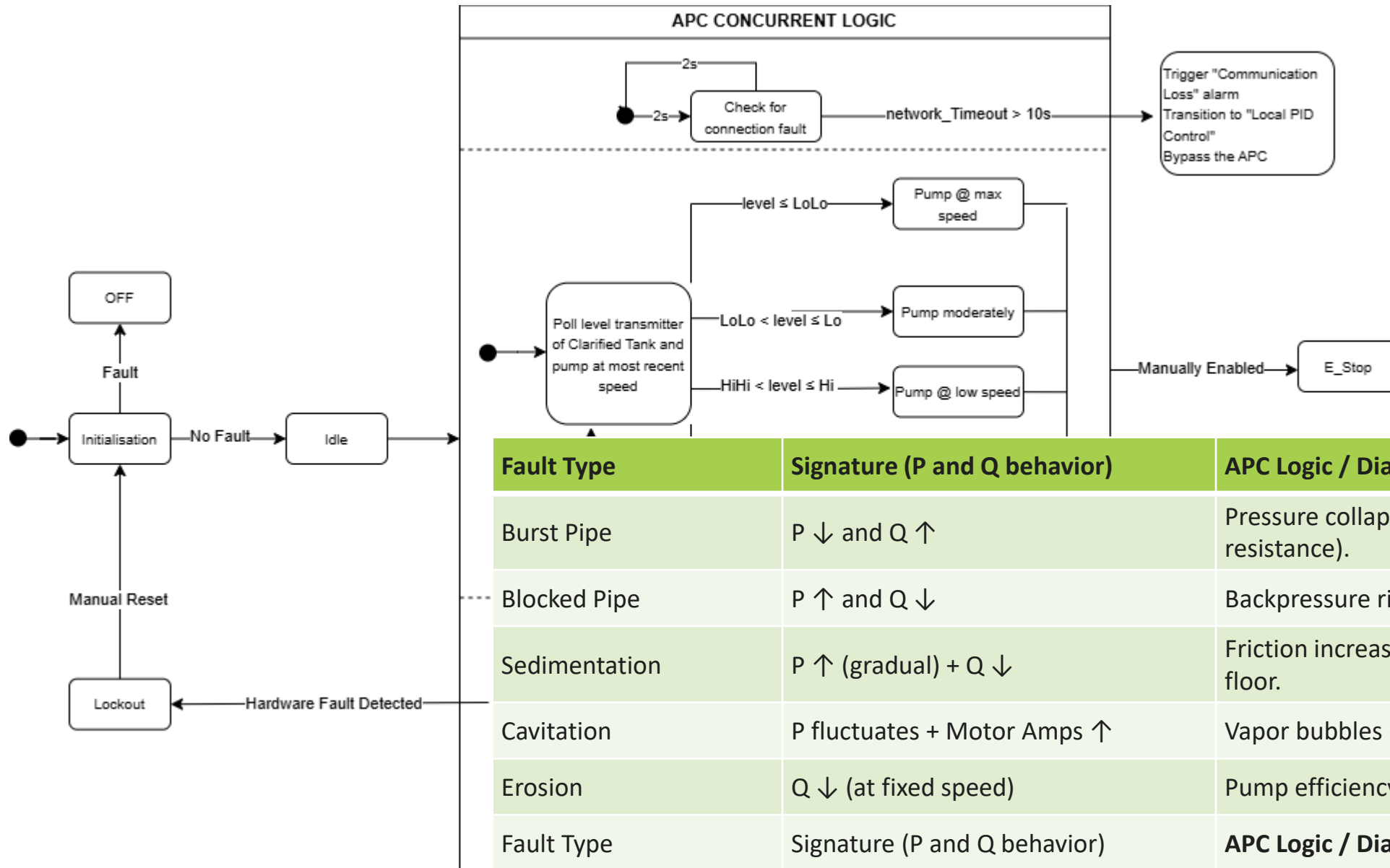


GG5_Conveyor_Window

SCADA only issues and receives commands and its control capabilities are confined to a series of if statements. It doesn't have the mathematical modelling capabilities that an APC has.

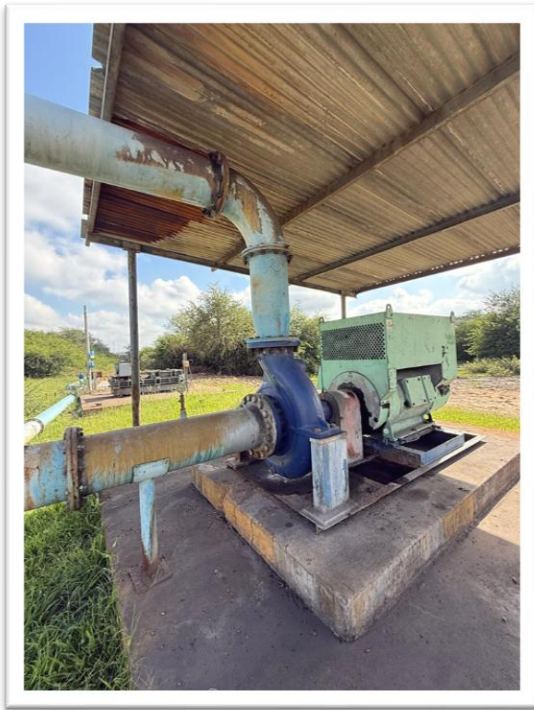
APC Control Logic





Recommendations

1. Installation of Suction Valves at the intake of pumps



Blue Pipeline B3 Pump

2. Incorporation of VSDs and APC to future pit dewatering projects



APC Operator screen & VSD

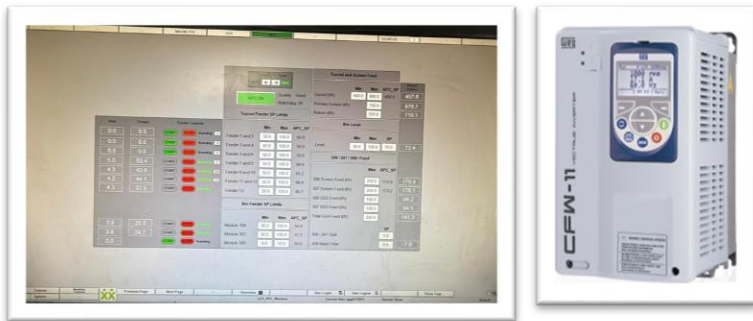
3. Ensure that pipes are at the regulated distance of 10m from B6 face



Blue Pipeline on B6

Consolidation

2. Incorporation of VSDs and APC to future pit dewatering projects



APC Operator screen & VSD

Cost

- Section 12L tax rebate
- Compliance with the Climate Change Act

Production

- ISO 50001 Certification
- Grow our clientele

Safety

- Reduced Scope 2 emissions
- Retained Asset Life
- Aligns with the plan to achieve carbon neutrality by 2050

Technical Growth

Designing for the Mine Value Chain
Fluid Mechanics Exposure

Professional Growth

Cross-Departmental Relationship Building
The importance of asking questions

Thank You

